

Summary

- ECE final-year student specializing in embedded systems, custom hardware design, and scalable cloud IoT architectures, with active research in renewable energy.

Education

- **Dr. B. C. Roy Engineering College** Durgapur, West Bengal
B.Tech in Electronics & Communication Engineering (ECE) *Expected Jun. 2026*

Achievements

- **PGET-2026 – Rank 35 in ECE** Jun. 2026
Post-Graduate Entrance Test (MAKAUT) – Top Performer

Technical Skills

- **Languages:** C/C++, Python, JavaScript/TypeScript, SQL
- **Embedded Systems:** Firmware Development, RTOS, Microcontrollers, IoT Protocols
- **Hardware Design:** PCB Layout, Circuit Analysis, Rapid Prototyping, Lab Instrumentation
- **Software & Cloud:** Full-Stack Web Development, REST APIs, Cloud Platforms, Containerization
- **Domain Expertise:** Renewable Energy Systems, Smart Grids, Telecom Networking (5G/FTTx)

Experience

- **BSNL** Burdwan, West Bengal
Telecommunications Intern *Summer 2025*
 - **Network Analysis:** Analyzed live telecom routing protocols and resolved fiber optic signal integrity issues.
- **Tata Power SDI (PMSGMBY)** Dhanbad, Jharkhand
Green Energy Supervisor Solar (NSQF Level 5, RPL Certified) *Winter 2026*
 - **RPL Assessment:** Supervised solar PV installations and site surveys; achieved NSQF Level 5 certification.

Projects

- **800W Synchronous MPPT Controller (ESP32)** Academic
ESP32 • MPPT • Incremental Conductance *Winter 2026*
 - **Overview:** Developed an 800W synchronous solar charge controller for rural electrification, maximizing conversion efficiency via INC algorithm.
- **Intelligent Smart Grid & Energy Telemetry Platform** Academic
Edge Computing • Power Metrology • Cloud DB • Web *Fall 2025*
 - **Overview:** Built a prepaid energy meter featuring real-time cloud telemetry, responsive dashboard, and sub-200ms relay cutoff.

Research & Publications

- **Weather-Integrated PEMS for Residential Hybrid Solar** Electrica
Under Review – electr-2025-0424 *Winter 2025*
 - **Overview:** Proposed a predictive management model utilizing weather data and smart metering to optimize hybrid solar scheduling.
- **ESP32-Based Synchronous MPPT SCC with INC Algorithm** ICICET 2026
Accepted – Speaker Category, Track 7 *Aug. 2026*
 - **Overview:** Presented findings on high-efficiency synchronous MPPT solutions tailored for rural electrification.

Languages

- **Bengali:** Native (C2) **Hindi:** Advanced (B2) **English:** Advanced (B2)

Interests

- Open-source hardware, DIY electronics, renewable energy advocacy, and audio storytelling.